LINK TO TINKERCARD :

https://www.tinkercad.com/things/i2I4QSWnTir-epic-jaban-borwo?sharecode=nLT6s1fSVI2SGHOguA-oZXUPM3ForNrXsn-amfISkHY

```
const int tempPin = A0; const
int ledPin = 8;
const int motorPin = 9;

void setup() {
pinMode(ledPin, OUTPUT);
  pinMode(motorPin, OUTPUT);

  Serial.begin(9600);
  Serial.println("CPS Monitoring and Control System");
  Serial.println("--------------------------------");
}

void loop() {   int sensorValue =
analogRead(tempPin);

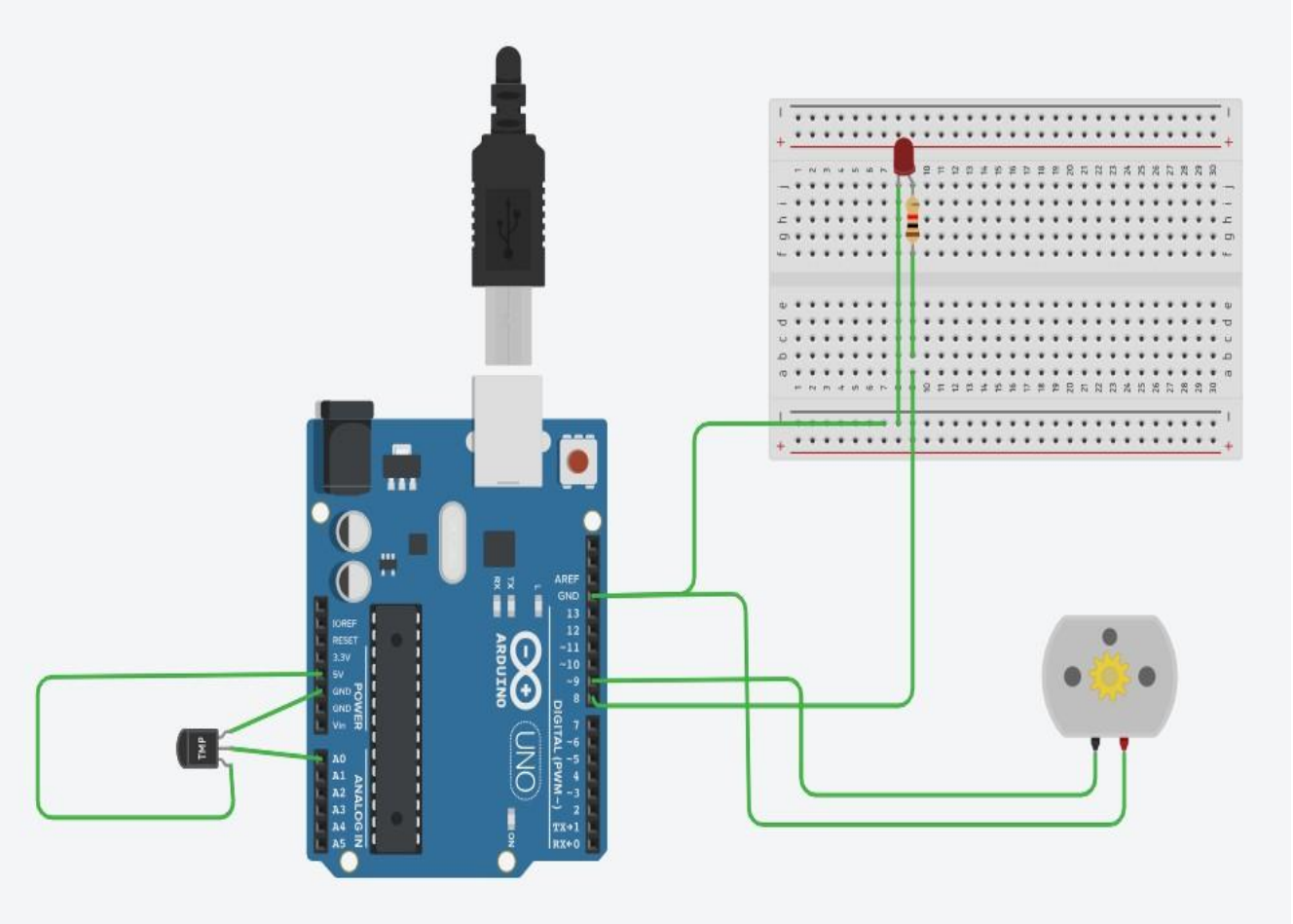
  float voltage = sensorValue * (5.0 / 1023.0);
float temperature = (voltage - 0.5) * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature >= 30.0) {
digitalWrite(ledPin, HIGH);
digitalWrite(motorPin, HIGH);
    Serial.println("High Temperature - Motor ON");
  } else {
    digitalWrite(ledPin, LOW);
digitalWrite(motorPin, LOW);
    Serial.println("Normal Temperature - Motor OFF");
  }

  Serial.println("--------------------------------");
```

```
  delay(1000);
}
```


AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
ARDUINO
UNO
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
TMP